



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

PROJECT 1

SECI1143: PROBABILITY & STATISTICAL DATA ANALYSIS 2022/2023 – SEMESTER 2

SOCIAL MEDIA AND ITS INFLUENCE IN OUR DAILY LIFE

LECTURER : DR NOORFA HASZLINNA BINTI
MUSTAFA
SECTION : 03
NAME : LIM JING YONG
MARCUS JOEY SAYNER
SOH HAN WEI

CONTENTS

1.0	Introduction	1
2.0	Data Collection	2
3.0	Data Analysis	5
3.1	Respondents' Biography	5
3.1.1	Respondents' Age	5
3.1.2	Respondents' Gender	6
3.1.3	Respondents' Occupation	7
3.2	Respondents' Usage on Social Media Applications	8
3.2.1	Type of Social Media Used	8
3.2.2	Respondents' Hours Spent on Phone Everyday	9
3.2.3	Respondents' Sleep Time	10
3.3	Respondents' Opinion on Social Media Consumption (General)	11
3.3.1	Do Respondents use Social Media for Entertainment?	11
3.3.2	Do Respondents think Social Media is Beneficial?	12
3.3.3	Do Respondents think Social Media is Harmful?	13
3.4	Respondent's Experience with respective Social Media	14
3.4.1	Reason of using Social Media	14
3.4.2	Frequency of Opening Respective Social Media in a day	18
3.4.3	Do Respondents enjoy using respective Social Media?	24
4.0	Conclusion	27

LIST OF TABLES

Table 2.1	List of Question Asked in Questionnaire	2-4
Table 3.1.1.1	Frequency Distribution of Age of Respondent	5
Table 3.1.1.2	Measure of Central Tendency of Age of Respondent	6
Table 3.1.2.1	Frequency and Percentage of Gender of Respondent	6
Table 3.1.3.1	Frequency Distribution of Occupation of Respondent	7
Table 3.2.1.1	Frequency Distribution of Usage of Social Media by Respondent	8
Table 3.2.2.1	Frequency and Percentage of Hours Respondent Spent on Phone (Range)	9
Table 3.3.1.1	Frequency of Respondent's Answer (Entertainment)	11
Table 3.3.2.1	Likert Scale Frequency on whether Social Media is Beneficial	12
Table 3.3.3.1	Likert Scale Frequency on whether Social Media is Harmful	13
Table 3.4.1.1	Respondent's Reason for Using Each Social Media Application	14
Table 3.4.2.1	Times Respondents have opened respective Social Media Application	18
Table 3.4.2.2	Measure of Quartile for Boxplot of Frequency of Tiktok Usage	19
Table 3.4.2.3	Measure of Quartile for Boxplot of Frequency of Twitter Usage	20

Table 3.4.2.4	Times Respondents have opened respective Social Media Application (Range)	21
Table 3.4.3.1	Respondent's Answer (Enjoyment) in Frequency and Percentage	24

LIST OF FIGURES

Figure 3.1.1.1	Dot Plot of Age of Respondent	5
Figure 3.1.2.1	Pie Chart of Gender of Respondent	6
Figure 3.1.3.1	Bar Chart of Occupation of Respondent	7
Figure 3.2.1.1	Bar Chart of Type of Social Media used among Respondents	8
Figure 3.2.2.1	Bar Chart of Daily Hours Spent by Respondents on Phone	9
Figure 3.2.3.1	Scatterplot of Respondents' Sleep Time and their Optimum Sleep Time	10
Figure 3.3.1.1	Pie Chart of Respondent's Answer (Entertainment)	11
Figure 3.3.2.1	Bar Chart of Likert Scale Frequency on whether Social Media is Beneficial	12
Figure 3.3.3.1	Bar Chart of Likert Scale Frequency on whether Social Media is Harmful	13
Figure 3.4.1.1	Bar Chart of Reason of using Youtube	15
Figure 3.4.1.2	Bar Chart of Reason of using Facebook	15
Figure 3.4.1.3	Bar Chart of Reason of using Instagram	15
Figure 3.4.1.4	Bar Chart of Reason of using Twitter	15
Figure 3.4.1.5	Bar Chart of Reason of using Tiktok	16
Figure 3.4.2.1	BoxPlot of Frequency of Tiktok Usage	19
Figure 3.4.2.2	BoxPlot of Frequency of Twitter Usage	19
Figure 3.4.2.3	Histogram of Frequency of Youtube Usage	20
Figure 3.4.2.4	Cumulative Frequency Curve for Instagram Usage	22
Figure 3.4.2.5	Cumulative Frequency Curve for Facebook Usage	22
Figure 3.4.3.1	Pie Chart of Enjoyment of using Youtube	24
Figure 3.4.3.2	Pie Chart of Enjoyment of using Facebook	24
Figure 3.4.3.3	Pie Chart of Enjoyment of using Instagram	24
Figure 3.4.3.4	Pie Chart of Enjoyment of using Twitter	24
Figure 3.4.3.5	Pie Chart of Enjoyment of using Tiktok	25

APPENDIX

Screenshot of Google Form	29
---------------------------	----

1.0 INTRODUCTION

Our project, titled “Social Media and its Influence in Our Daily Life”, is research aimed to investigate the impact of social media on undergraduates that use them every single day! As we all know, phones have basically become one of the necessities of human life, and such factors also apply to social media. It is rather abnormal to see someone in this generation completely out of the loop of social media’s vitality. Whether it is for entertainment, education, connectivity, or even unhealthy purposes, everyone has their own way of using social media. Hence, we are conducting this research to collect data on elements surrounding the effect social media brings about among us undergraduates.

In this project, we will be questioning undergraduates, specifically from the Faculty of Computing, University of Technology Malaysia (UTM) about how heavy the presence of social media is on a daily basis. We want to find out how many types of social media they use, how many hours they spend on it, and their personal opinions on whether social media is truly beneficial or disadvantageous to them. There exists multiple social media that are widely used among the population, and the scope of social media included in our research is as such: Youtube, Facebook, Instagram, Twitter, and Tiktok.

The questionnaire we prepared, in Google Form, is distributed to undergraduates of the Faculty of Computing online, through media such as Telegram and Whatsapp, and we have successfully gathered 120 responses from the students of UTM. The expectation we have for this research is that social media’s impact will be more towards profit than loss in terms of its effects on undergraduates, as it serves as one of the crucial tools for socializing to expand their friend circle, entertainment to relieve stress, and gaining knowledge where sometimes only social media is capable of giving.

2.0 DATA COLLECTION

The methodology that we used in this project to collect data is through conducting a survey via Google Form. The survey consists of some parts, which are the personal information of the respondent, the usage of social media by the respondent, opinion of respondent on social media and the usage of each social media by the respondent. Some of the questions in Google Form will be shown at the appendix part which is at the end of the project.

No	Question	Answer	Level of Measurement
1	Age	• Short Answer (Accept Integer Only)	Ratio
2	Gender	• Male • Female	Nominal
3	Current occupation	• Student • Worker • Unemployed	Nominal
4	Number of social media currently used	• Youtube • Facebook • Instagram • Twitter • Tiktok	Ratio
5	Total hours spent on phone on a daily basis	• 0 - 2 hours • 3 - 5 hours • 6 - 8 hours • 9 - 11 hours	Ratio
6	What time do you normally sleep?	• Time	Interval
7	In your opinion, when do you think is the most optimum time to sleep?	• Time	Interval
8	Do you use social media for entertainment purposes?	• Yes • No	Nominal

No	Question	Answer	Level of Measurement
9	Do you agree that social media is beneficial to people?	1 (Strongly agree) - 5 (Strongly disagree)	Ordinal
10	Do you agree that social media is harmful to people?	1 (Strongly agree) - 5 (Strongly disagree)	Ordinal
11	Why do you use Youtube	- Entertainment - Education - Other	Nominal
12	How often do you open Youtube in one day	Short Answer	Ratio
13	Do you enjoy using Youtube	- Yes - No	Nominal
14	Why do you use Facebook	- Entertainment - Education - Other	Nominal
15	How often do you open Facebook in one day	Short Answer	Ratio
16	Do you enjoy using Facebook	- Yes - No	Nominal
17	Why do you use Instagram	- Entertainment - Education - Other	Nominal
18	How often do you open Instagram in one day	Short Answer	Ratio
19	Do you enjoy using Instagram	- Yes - No	Nominal

No	Question	Answer	Level of Measurement
20	Why do you use Twitter	• Entertainment • Education • Other	Nominal
21	How often do you open Twitter in one day	• Short Answer	Ratio
22	Do you enjoy using Twitter	• Yes • No	Nominal
23	Why do you use Tiktok	• Entertainment • Education • Other	Nominal
24	How often do you open Tiktok in one day	• Short Answer	Ratio
25	Do you enjoy using Tiktok	• Yes • No	Nominal

3.0 DATA ANALYSIS

3.1 RESPONDENT BIOGRAPHY

3.1.1 Respondents' Age

Age	Frequency
16	2
18	2
19	16
20	44
21	39
22	8
23	7
24	2
TOTAL	120

Table 3.1.1.1 Frequency Distribution of Age of Respondent

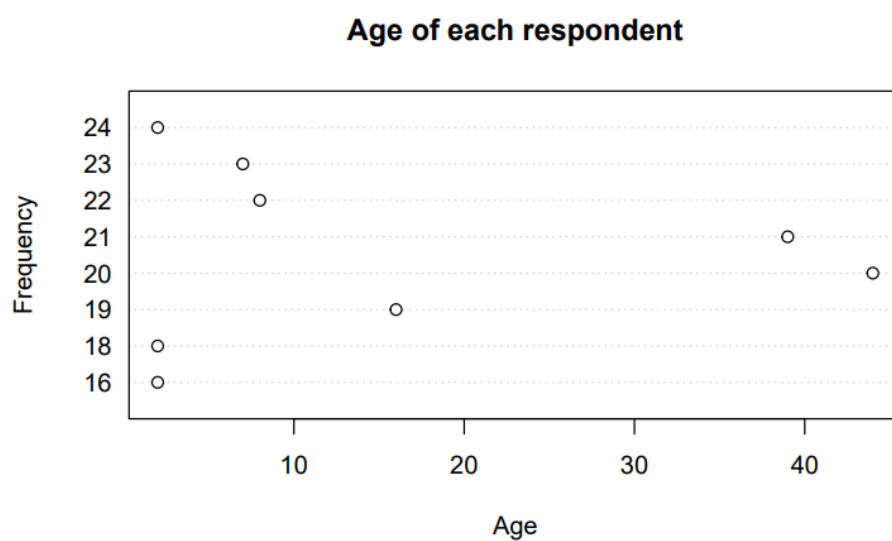


Figure 3.1.1.1 Dot Plot of Age of respondent

Measure of Central Tendency	Value
Mean	20.4667
Median	20
Mode	20

Table 3.1.1.2 Measure of Central Tendency of Age of Respondents

As seen in Table 3.1.1.1 and Figure 3.1.1.1, the age distribution of respondents that have answered our questionnaire are around the range of age 16 to 24. Since this research is focused on undergraduates of the Faculty of Computing, UTM, most of the respondents are of age 20, with the frequency of 44. Students of age 21 come close with the frequency of 39, and there are 16 19-year-old respondents. There are also 8 22-year-old and 7 23-year old respondents, and the rest, of age 16, 18, and 24, hold a frequency of 2 students. Additionally, it can be seen from Table 3.1.1.2 that the data we gathered will mostly be from respondents of age 20, as our median and mode both are 20, and mean being 20.4667.

3.1.2 Respondents' Gender

Gender	Frequency	Percentage
Male	60	50%
Female	60	50%
TOTAL	120	100%

Table 3.1.2.1 Frequency and Percentage of Gender of Respondent

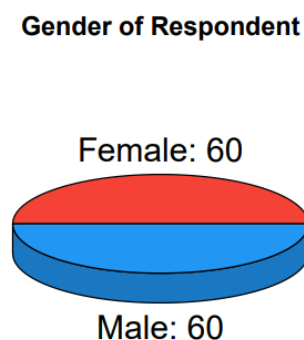


Figure 3.1.2.1 Pie Chart of Gender of Respondent

As indicated in Table 3.1.2.1 and Figure 3.1.2.1, which are the frequency table and pie chart of respondent gender, the frequency of male and female is 60. As a result, both will have the same relative frequency, 0.5, as well as the same proportion, 50%. There are 120 respondents of each gender. In addition, the pie chart of responder gender illustrates that red means female and blue means male.

3.1.3 Respondents' Occupation

Occupation	Frequency
Student	117
Unemployed	2
Worker	1
TOTAL	120

Table 3.1.3.1 Frequency Distribution of Occupation of Respondent

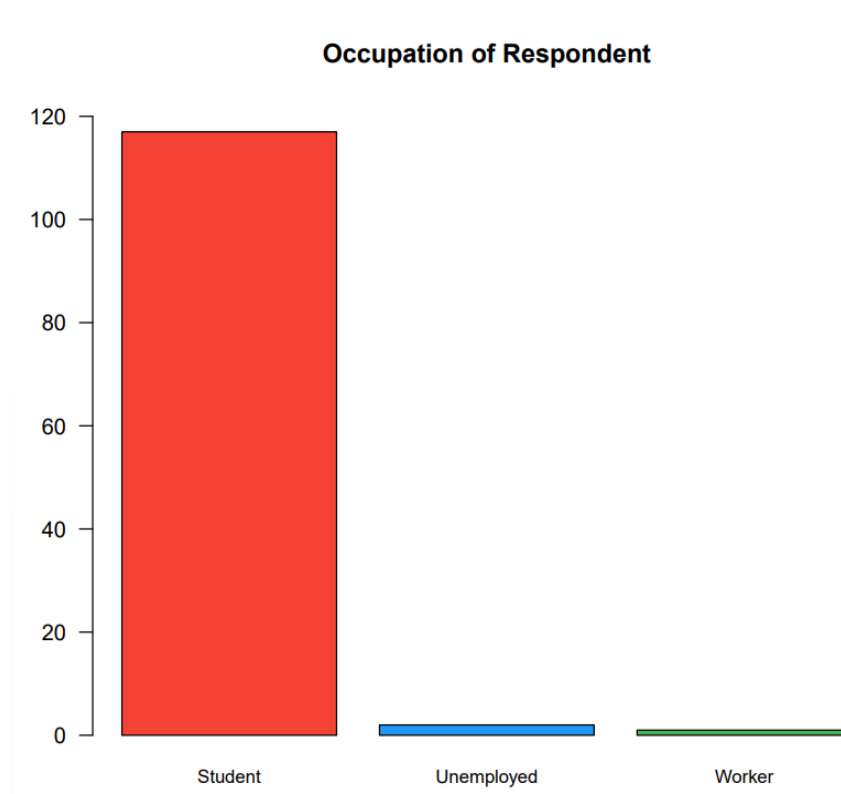


Figure 3.1.3.1 Bar Chart of Occupation of Respondent

As stated in Table 3.1.3.1 and Figure 3.1.3.1, the respondents' occupations in the University Technology Malaysia, might be student, unemployed, or worker. The occupation with the most respondents is Student studying at the faculty, with 117 respondents. Following that, it will be the unemployed faculty member, who will have only two respondents. Finally, only one respondent's occupation was recorded in the data which is the lowest among all the three occupations. We can conclude that 117 out of a total of 120 respondents are students, as well as the mode of occupation.

3.2 Respondents' Usage on Social Media Applications

3.2.1 Type of Social Media Used

Type of Social Media	Frequency
Facebook	103
Instagram	114
Tiktok	42
Twitter	32
Youtube	110

Table 3.2.1.1 Frequency Distribution of Usage of Social Media by Respondent

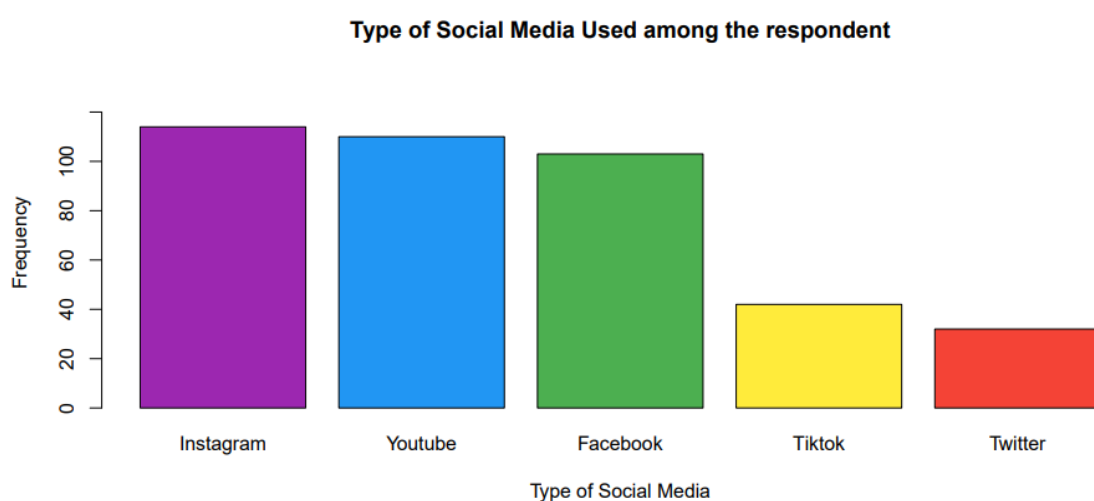


Figure 3.2.1.1 Bar Chart of Social Media used among Respondents

From Table 3.2.1.1 and Figure 3.2.1.1, we can see the frequency of each social media being used by our respondents. The total frequency can exceed 120 as one respondent is allowed to choose more than one social media that they use. Instagram has the highest frequency of 114, which means almost all undergraduates are currently using it. The same case goes to Facebook and Youtube, where both social media has a frequency of 103 and 110 respectively. For Twitter and Tiktok however, these social media only have 42 and 32 users from our sample respectively. This means our sample of population prefers using Facebook, Instagram, and Youtube much more than Tiktok and Twitter.

3.2.2 Respondents' Hours Spent on Phone Everyday

Range (Hours)	Frequency	Percentage
0 - 2	15	12.5%
3 - 5	44	36.67%
6 - 8	41	34.17%
9 - 11	20	16.67%
TOTAL	120	100%

Table 3.2.2.1 Frequency and Percentage of Hours Respondent Spent on Phone (Range)

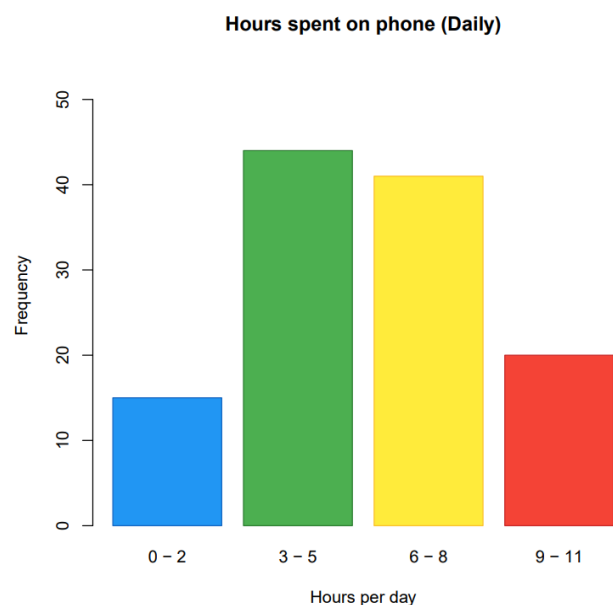


Figure 3.2.2.1 Bar Chart of Daily Hours spent by Respondent on Phone

In Table 3.2.2.1 and Figure 3.2.2.1, the hours spent by respondents on phone everyday are grouped into 4 parts, each bar ranging in 2 hours. 44 and 41 respondents open social media 3 to 5 times and 6 to 8 times everyday. 15 respondents only open social media 0 to 2 times, and 20 respondents excessively use social media, up to 9 to 11 times. In percentages, respondents mostly are consuming social media in the normal amount. This means social media consumption, if not at all, barely has any negative effect on student's studies. This might not apply to the students who use social media 9 to 11 times as it might be too excessive for a student to use social media this frequently in a day.

3.2.3 Respondents' Sleep Time

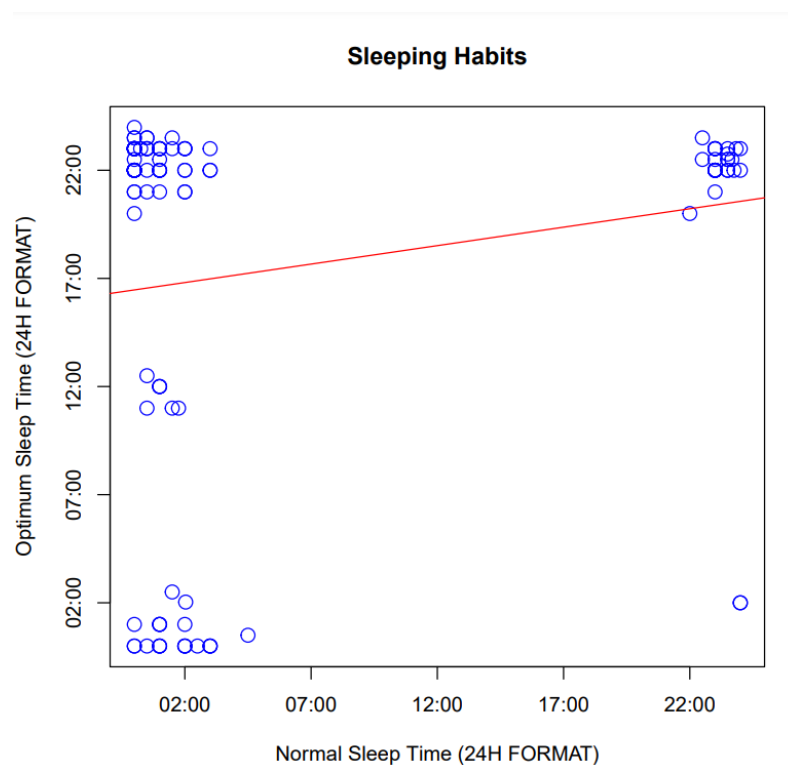


Figure 3.2.3 Scatterplot of Respondents' Sleep Time and their Optimum Sleep Time

According to Figure 3.2.3, the scatterplot describes students' time to sleep and their opinion on the optimum time to sleep. Our initial expectation of this chart would have most dots close to 10 p.m, but surprisingly most of the responses are close to around 12.00 am. This could be heavily impacted by the existence of social media, as it is something fun but time consuming. Students might lose time of track due to usage of social media, hence why most of the dots are close to 12.00 am. It can be concluded that social media applications might have impacted undergraduate students to sleep slightly later than usual.

3.3 Respondents' Opinion on Social Media Consumption (General)

3.3.1 Do respondents use social media for entertainment?

Answer	Frequency
Yes	115
No	5
TOTAL	120

Table 3.3.1.1 Frequency of Respondent's Answer (Entertainment)

Usage of social media for entertainment purpose

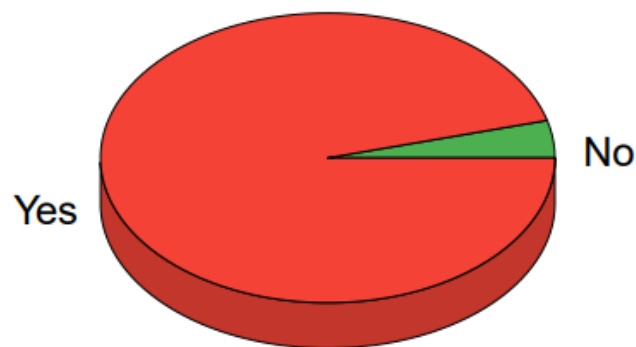


Figure 3.3.1.1 Pie Chart of Respondent's Answer (Entertainment)

Table 3.3.1.1 and Figure 3.3.1.1 show that it is about using social media for amusement reasons. It demonstrates that a large number of respondents, approximately 115 out of 120 respondents, use social media as an entertainment medium. The percentage of respondents who use social media for entertainment purposes is 95.83%, whereas the percentage of respondents who use it for other purposes is 4.17%, or around 5 out of 120 respondents. It is plainly stated that respondents utilise social media for recreational purposes in their spare time. Furthermore, the pie chart shows that the colour red is considered "Yes," while the colour green is considered "No."

3.3.2 Do Respondents think Social Media is Beneficial?

Likert Scale	Frequency
Strongly Agree	37
Agree	63
Neutral	18
Disagree	1
Strongly Disagree	1
TOTAL	120

Table 3.2.2.1 Likert Scale Frequency on whether Social Media is Beneficial

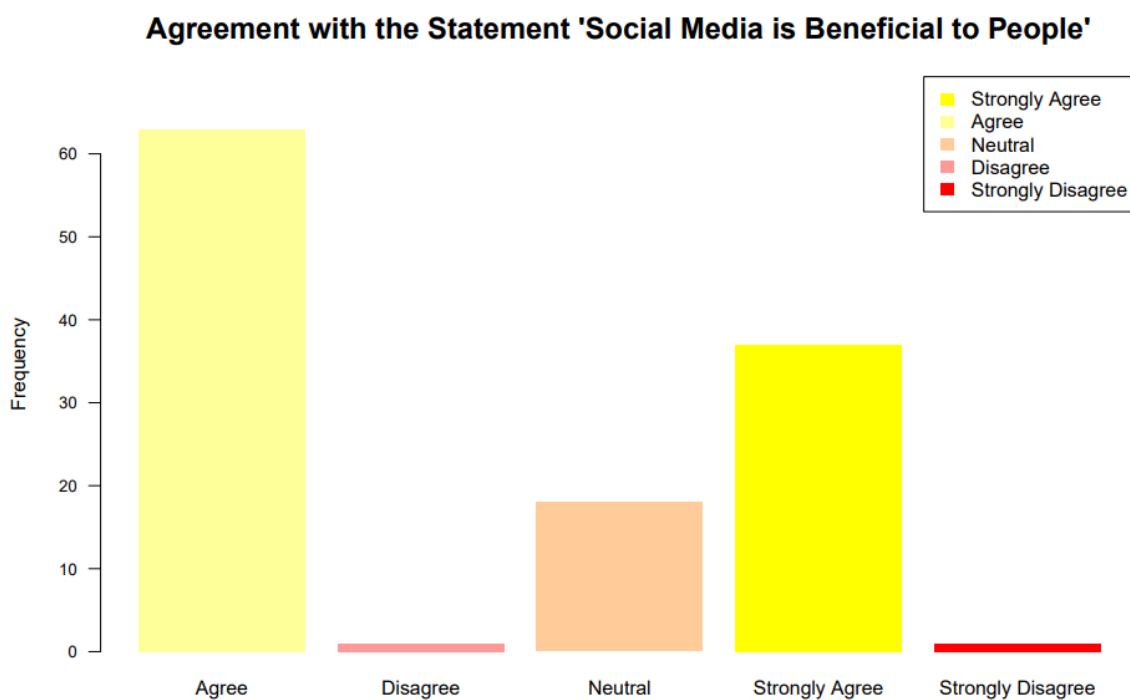


Figure 3.3.2.1 Bar Chart of Likert Scale Frequency on whether Social Media is Beneficial

Table 3.3.2.1 and Figure 3.3.2.1 depicts undergraduates' opinions on whether social media is beneficial to people. 37 respondents strongly agree, 63 respondents agree, 18 respondents are neutral about it, and only 1 respondents strongly disagree and disagree respectively. We can tell from the bar chart that there exists quite a unionized opinion that

social media is indeed beneficial to most people. This can be due to the fact that social media is one of the important sources of entertainment for university students to relieve stress. While overuse of social media could lead to opposite effects, responsibly using social media does become helpful for undergraduates to find fun in life.

3.3.3 Do Respondents think Social Media is Harmful?

Likert Scale	Frequency
Strongly Agree	14
Agree	42
Neutral	56
Disagree	6
Strongly Disagree	2
TOTAL	120

Table 3.3.3.1 Likert Scale Frequency on whether Social Media is Harmful

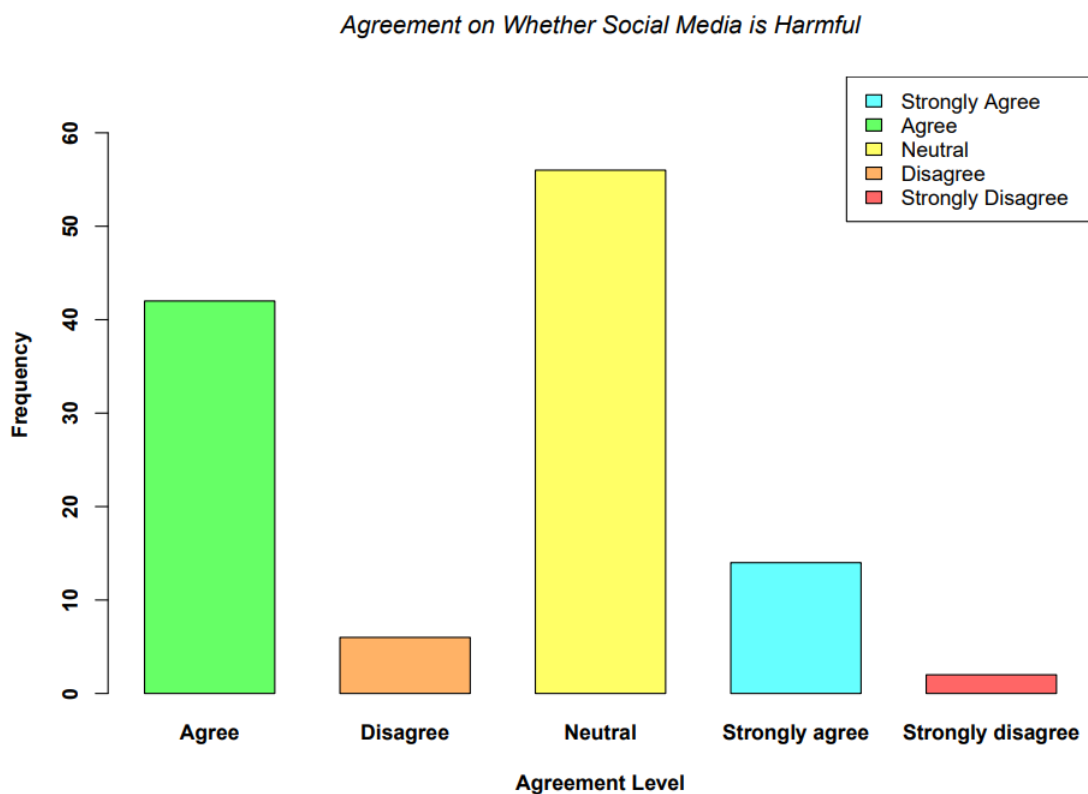


Figure 3.3.3.1 Bar Chart of Likert Scale Frequency on whether Social Media is Harmful

It is about whether respondents agree that social media is harmful, as shown in Table 3.3.3.1 and Figure 3.3.3.1. In this bar graph, it shows that the x-axis is the agreement level and the y-axis is the frequency of the respondents that we collected. With 56 out of 120 respondents, it is most likely that they are neutral regarding social media being harmful. Furthermore, 42 respondents believe that social media is harmful to us. Furthermore, 14 out of 120 respondents strongly agree that social media is harmful. Furthermore, data collected about 6 over 120 respondents disagree on whether social media is harmful. Finally, the lowest number of responders will strongly disagree, with only two persons disagreeing. In conclusion, the majority of respondents are neutral about it and prefer to maintain control over social media.

3.4 Respondent's Experience with respective Social Media

3.4.1 Reason of using Social Media

Type of Social Media	Frequency for each Reason		
	Education	Entertainment	Other
Youtube	91	114	18
Facebook	33	103	26
Instagram	29	112	23
Twitter	17	46	81
Tiktok	16	54	74
TOTAL	186	429	222

Table 3.4.1.1 Respondent's Reason for Using Each Social Media Application

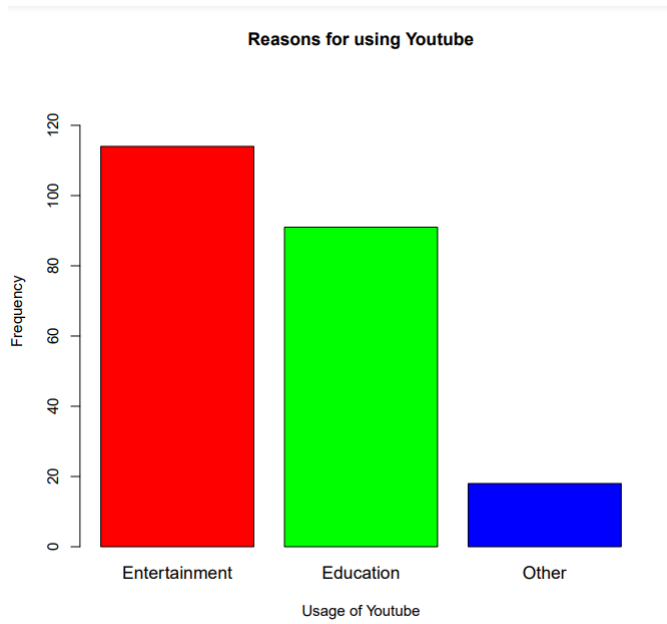


Figure 3.4.1.1 Bar Chart of Reason of using Youtube

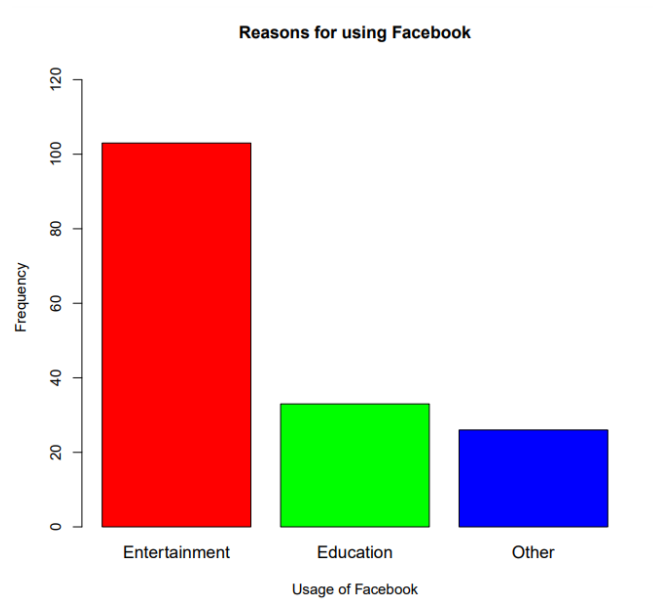


Figure 3.4.1.2 Bar Chart of Reason of using Facebook

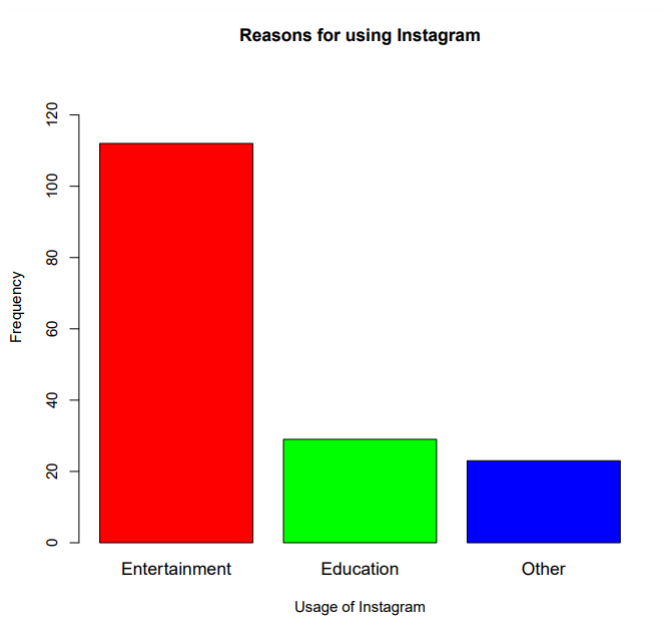


Figure 3.4.1.3 Bar Chart of Reason of using Instagram

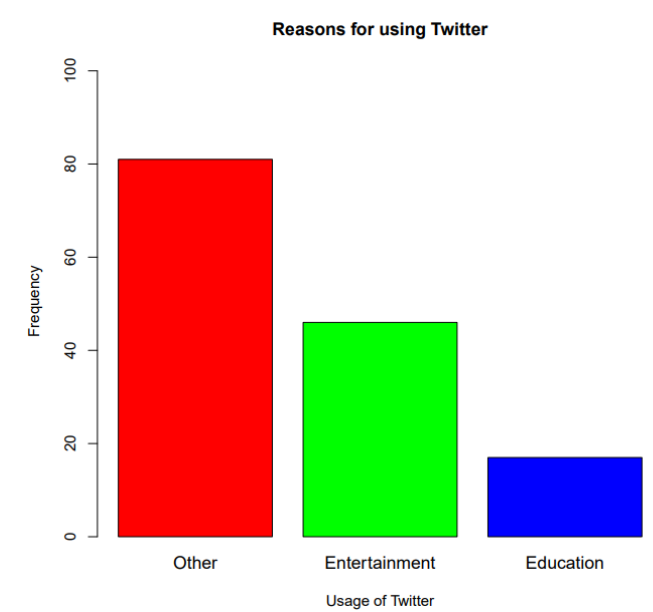


Figure 3.4.1.4 Bar Chart of Reason of using Twitter

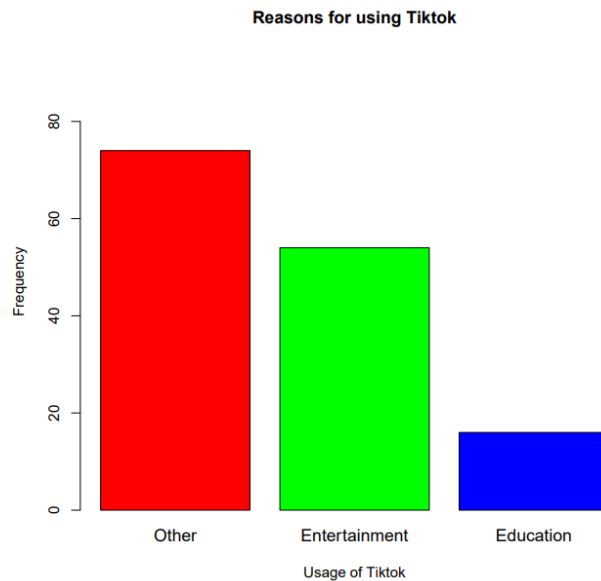


Figure 3.4.1.5 Bar Chart of Reason of Using Tiktok

As indicated in Table 3.4.1.1 and all five figures, which are Figures 3.4.1.1, 3.4.1.2, 3.4.1.3, 3.4.1.4, and 3.4.1.5. The five statistics are divided into five social media platforms: YouTube, Facebook, Instagram, Twitter, and Tiktok. We believe that valid reasons for respondents' use of social media are for entertainment, education, and other purposes. In addition, respondents have the option of choosing a few different reasons on each social media platform. In the bar chart, we can see that the highest frequency is usually red, while the lowest frequency is blue.

First and foremost, we shall investigate the reasons for which YouTube is used for educational, entertainment, and other objectives. With the mode of reasons for using YouTube, 114 responses said that it is for watching entertainment videos. Furthermore, 91 responses use YouTube for educational and research purposes. Finally, with roughly 18 responses, other objectives receive the fewest responses. To summarise, YouTube is a wonderful platform to explore new things and entertain ourselves in our spare time.

The second figure depicts the reasons for using Facebook, which could be educational, entertaining, and other purposes. The most common reason respondents use Facebook is for educational purposes, with 103 responses, followed by for educational purposes, with 33 responses. At last, with 26 reasons, the lowest frequency of reasons for using Facebook is for other purposes. We can see that respondents at our university use Facebook for entertainment, education, and, finally, other purposes.

The third figure looks into the reasons respondents use Instagram, one of the most popular social media platforms. As is customary, the reasons could be educational, entertaining, and other purposes. It is used for entertainment, according to 112 responses. Furthermore, it has around 29 responses for educational purposes, followed by other purposes, which has 23 responses with the lowest frequency of reasons for using Instagram. In a nutshell, respondents use Instagram primarily for entertainment purposes.

Furthermore, the fourth figure will provide insight into why people use Twitter as a social media platform for educational, entertainment, and other reasons. With 81 responses, the most common reason is that respondents use Twitter for purposes other than entertainment or education. With 46 responses, the reason for entertainment came in second. Last but not least, with only 17 responses, the lowest frequency of reasons for using Twitter is for educational purposes. As a recap, Twitter is a platform that is used for a variety of purposes, including the dissemination of world news.

The reasons for using Tiktok, a popular social media platform among this generation, are depicted in the final figure. Respondents use Tiktok for educational purposes the most, with 74 responses, out of the other two reasons combined. Furthermore, the median for those three reasons for using Tiktok, which has around 54 responses, is for entertainment purposes. With 16 responses, the lowest frequency will be for educational purposes. In summary, Tiktok is a fantastic platform for sharing information all over the world, not just educational and entertaining shorts.

In general, entertainment purposes have the highest frequency of responses among all social media platforms, with 429 responses. With approximately 222 responses, the median of overall social media reasons will be for other purposes. Furthermore, the lowest overall frequency will be for educational purposes. It was clearly stated that those social media platforms usually serve the purpose for entertainment and are rarely used for educational purposes.

3.4.2 Frequency of Opening Respective Social Media in a day

Times of Opening Social Media	Frequency of opening Social Media (daily)		
	Youtube	Twitter	Tiktok
0	2	56	22
1	19	6	24
2	15	14	31
3	21	8	10
4	20	7	12
5	10	9	4
6	12	9	6
7	9	3	6
8	7	4	2
9	3	1	2
10	1	2	0
11	0	1	0
12	1	0	1
TOTAL	120	120	120

Table 3.4.2.1 Times Respondents have opened respective Social Media Application

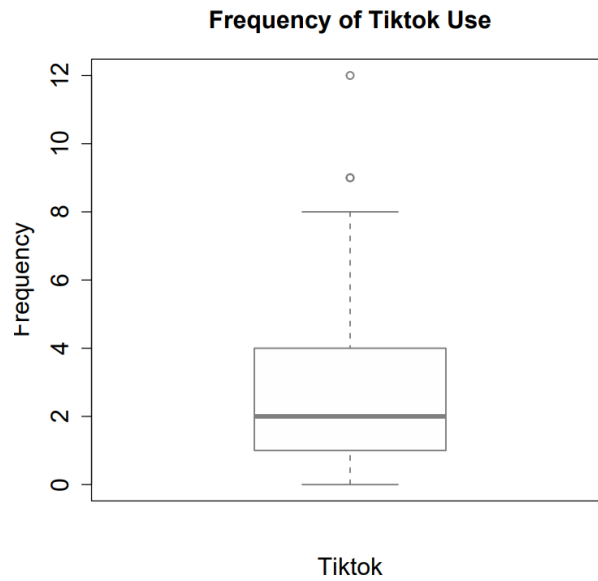


Figure 3.4.2.1 BoxPlot of Frequency of Tiktok Usage

Quartile	Frequency
0.25	1.0
0.50	2.0
0.75	4.0

Table 3.4.2.2 Measure of Quartile for Boxplot of Frequency of Tiktok Usage

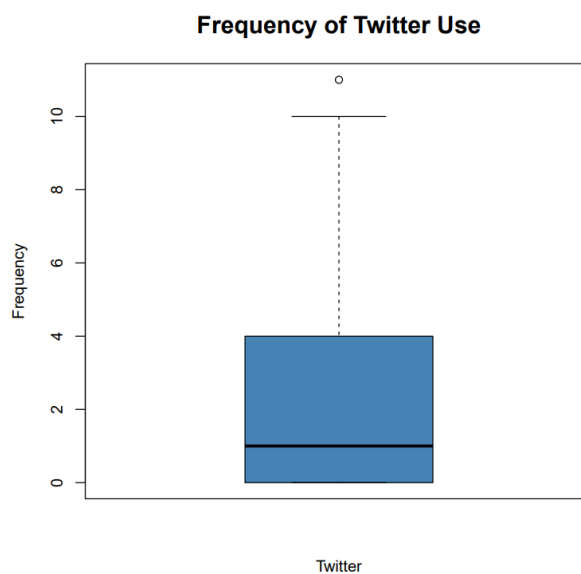


Figure 3.4.2.2 BoxPlot of Frequency of Twitter Usage

Quartile	Frequency
0.25	0.0
0.50	1.0
0.75	4.0

Table 3.4.2.3 Measure of Quartile for Boxplot of Frequency of Twitter Usage

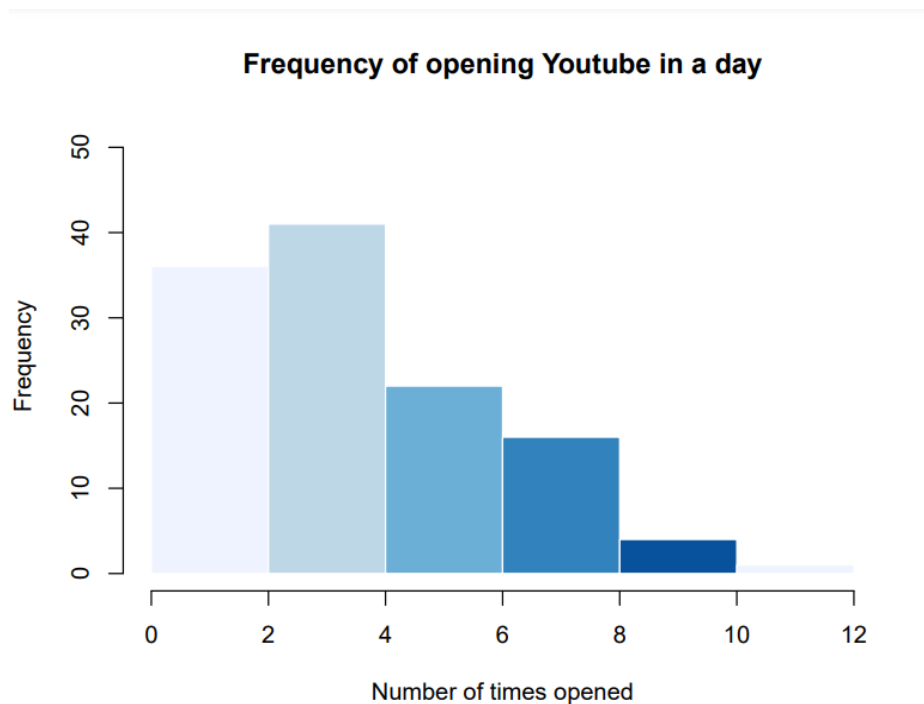


Figure 3.4.2.3 Histogram of Frequency of Youtube Usage

In Table 3.4.2.1, the frequency of undergraduates using specific social media, namely Youtube, Twitter, and Tiktok, in a day is distributed from 0 to 12 times. We have displayed said data into box plots and histogram which is depicted in Figure 3.4.2.1, Figure 3.4.2.2, and Figure 3.4.2.3.

First, in Figure 3.4.2.1, we can see that our box is slightly skewed to the bottom, indicating that more students only use Tiktok around 1 to 4 times per day. In table 3.4.2.2 our first quartile lies on 1.0, second quartile lies on 2.0, and third quartile lies on 4.0. There also exist two outliers, with the frequency of 9 and 12, that use Tiktok very frequently. This data tells us that most students rarely use Tiktok as they might prefer the other social media mentioned in this research.

Additionally, compared to Tiktok, Twitter has even fewer users among the respondents. The box is extremely close to the value of 0, having the first quartile of 0.0, second quartile of 1.0, and third quartile of 4.0. It seems that not a lot of undergraduates use Twitter regularly except the only outlier of 11 times. With that, we can conclude that Twitter is what most students are currently using as their main social media.

Last but not least, Figure 3.4.2.3 explains the frequency of students using Youtube in Histogram form. We can see that the highest number of times undergraduates open Youtube is 2 to 4 times. Compared to Tiktok and Twitter, Youtube has a larger audience, due to the fact that Youtube could also be highly educational, being able to provide students with information or tutorials that they might need for their study.

Times of Opening Social Media (Range)	Type of Social Media			
	Instagram		Facebook	
	Frequency	Cumulative Frequency	Frequency	Cumulative Frequency
-0.5 - 1	8	8	34	34
1 - 2.5	9	17	17	51
2.5 - 4	29	46	19	70
4 - 5.5	5	51	11	81
5.5 - 7	44	95	29	110
7 - 8.5	17	112	5	115
8.5 - 10	6	118	4	119
10 - 11.5	0	118	0	119
11.5 - 13	2	120	1	120

Table 3.4.2.4 Times Respondents have opened respective Social Media Application (Range)

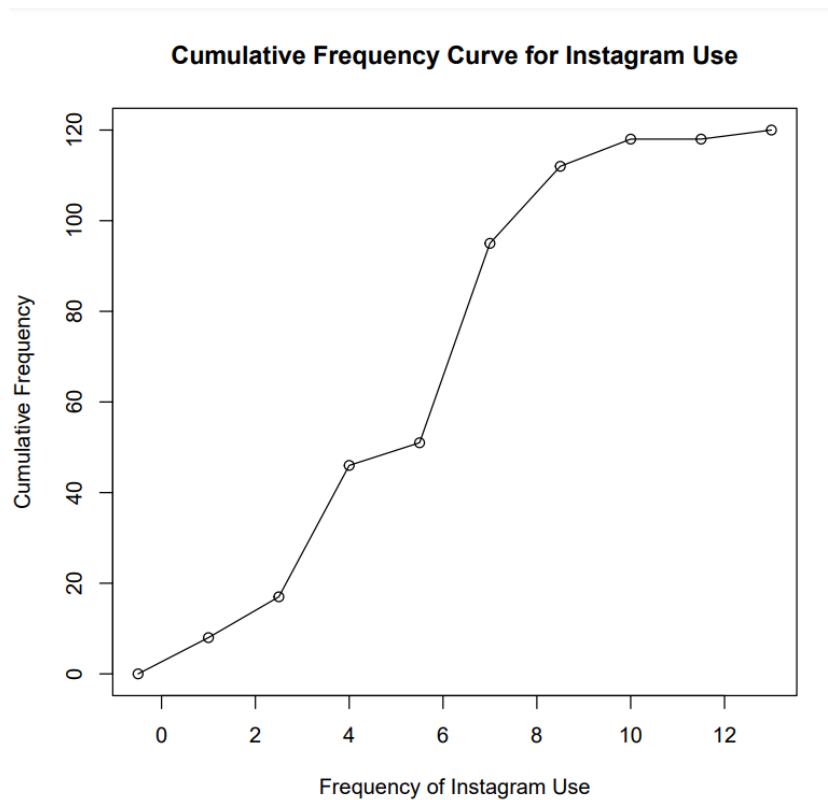


Figure 3.4.2.4 Cumulative Frequency Curve of Instagram Usage

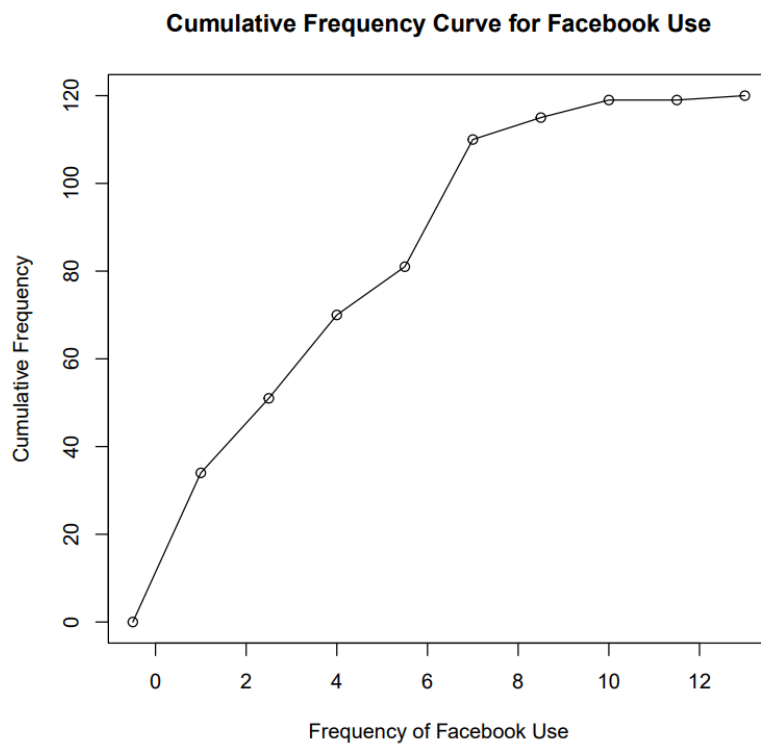


Figure 3.4.2.5 Cumulative Frequency Curve for Facebook Usage

In this table, Table 3.4.2.4, and both figures, Figures 3.4.2.4 and 3.4.2.5, we will investigate both social media platforms, Facebook and Instagram usage, which is defined as how many times respondents open those social media platforms.

Instagram is the most popular social media platform, with 44 respondents opening it between 5.5 and 7 times per day. The second highest will be 2.5 - 4 times, with approximately 29 over 120 respondents. Furthermore, with 17 over 120 respondents, respondents in third place visit Instagram approximately 7 - 8.5 times. Unfortunately, with 0 respondents, the lowest frequency that respondents answered is using Instagram for 10 - 11.5 times. Furthermore, the highest number of times opening Instagram, 11.5 - 13 times, has only 2 more than 120 respondents. To summarise, respondents use Instagram 5.5 - 7 times per week, which is still a neutral usage of the social media platform.

In comparison to other social media platforms, Facebook has the highest frequency of use, with 34 respondents using it for entertainment, education, and other purposes only once per day. This demonstrates that many teenagers in this generation did not use Facebook as their social media platform. Otherwise, there are still 29 over 120 respondents who are familiar with Facebook and use it as a leisure activity. With 0 respondents, the lowest number of times a respondent uses Facebook is 10 - 11.5 times per day, which is a lot of times opening a social media platform in a day. To break it down, people nowadays barely open social media platforms because they are too busy studying or working at university.

In comparison to both cumulative frequency curves, Figures 3.4.2.4 and 3.4.2.5, the peak of increment of Facebook usage is faster than the peak of increment of Instagram usage due to Facebook having higher frequency at the starting point, at -0.5 - 1 times of the cumulative frequency graph. Finally, it demonstrates that teenagers nowadays prefer Instagram to Facebook as their social media platform.

3.4.3 Do Respondents Enjoy using Respective Social Media?

Type of Social Media	Answers			
	Yes		No	
	Frequency	Percentage	Frequency	Percentage
Youtube	115	95.83%	5	4.17%
Facebook	98	81.67%	22	18.33%
Instagram	116	96.67%	4	3.33%
Twitter	43	35.83%	77	64.17%
Tiktok	52	43.33%	68	56.67%

Table 3.4.3.1 Respondent's Answer (Enjoyment) in Frequency and Percentage

Enjoyment of Using Youtube

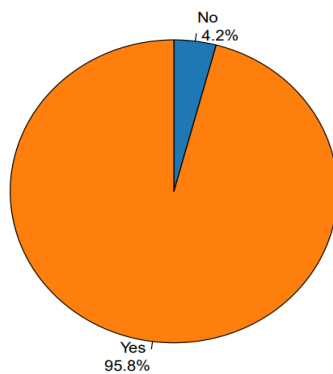


Figure 3.4.3.1 Pie Chart of Enjoyment of using Youtube

Enjoyment of Facebook Users

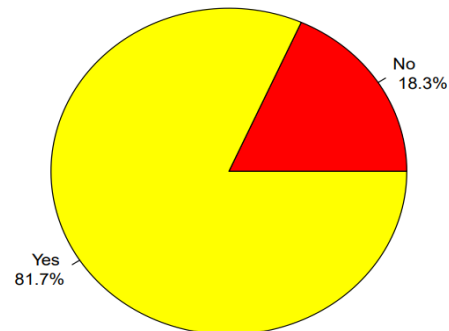


Figure 3.4.3.2 Pie Chart of Enjoyment of using Facebook

Enjoyment of Instagram Users

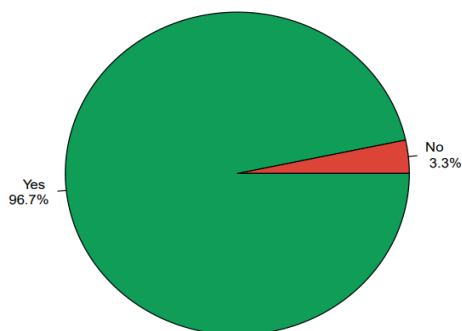


Figure 3.4.3.3 Pie Chart of Enjoyment of using Instagram

Enjoyment of Twitter Users

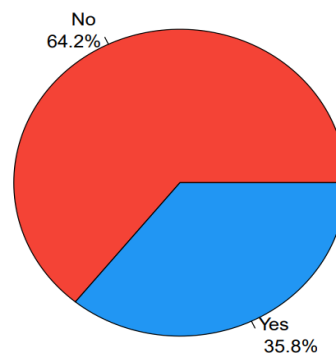


Figure 3.4.3.4 Pie Chart of Enjoyment of using Twitter

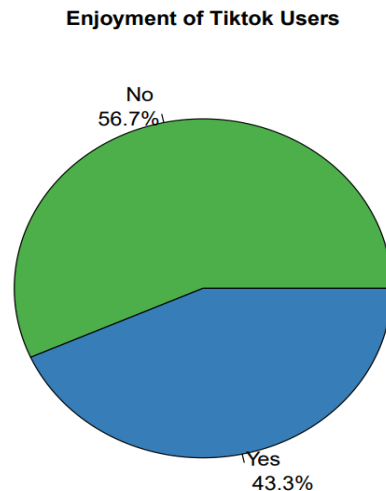


Figure 3.4.3.5 Pie Chart of Enjoyment of using Tiktok

In this table, Table 3.4.3.1, and five social media figures of pie chart, Figure 3.4.3.1, Figure 3.4.3.2, Figure 3.4.3.3, Figure 3.4.3.4, and Figure 3.4.3.5, will determine about respondents' enjoyment of using those social media platforms such as Youtube, Facebook, Instagram, Twitter, and Tiktok.

To begin with, approximately 115 out of 120 respondents enjoyed using YouTube as a social media platform, otherwise, only 5 respondents did not enjoy using Youtube. Figure 3.4.3.1, which is a pie chart that shows that the percentage of respondents who answer "Yes" is 95.83%, while the percentage of respondents who answer "No" because they do not enjoy it is 4.17%. To summarize, the highest frequency respondents' enjoyment is "Yes".

Furthermore, around 98 out of 120 respondents enjoy using Facebook as a social media platform. Otherwise, 22 respondents do not like Facebook as a social media platform. We can see in the pie chart, Figure 3.4.3.2, that the percentage of respondents who enjoy using Facebook is 81.67%. On the other hand, approximately 18.33% of respondents do not enjoy using Facebook. To break it down, "Yes" received the highest percentage of responses, indicating that respondents still enjoy Facebook.

In addition, in the table, we will investigate the enjoyment of using Instagram as a social media platform. It is clear that respondents enjoy using Instagram, with 116 out of 120 responding. Only four respondents, on the other hand, did not enjoy using Instagram as a social media platform. Figure 3.4.3.3 shows that the highest percentage of respondents who enjoy using Instagram is 96.67% in the pie chart. Otherwise, only 3.33% did not enjoy the

Instagram social media platform. To summarise, most respondents enjoyed using Instagram because it is a popular platform among this generation.

Furthermore, the table depicts the enjoyment of using the social media platform Twitter. Unfortunately, only 43 out of 120 respondents enjoy using Twitter as a social media platform, resulting in the lowest frequency of responding "Yes" among other social media platforms. Otherwise, the majority of respondents with 77 out of 120 will say "No" because they dislike Twitter. Figure 3.4.3.4 shows that the percentage of "No" (64.17%) outnumbers the percentage of "Yes" (35.83%). As a result, respondents in this generation did not enjoy using Twitter as their social media platform.

Finally, for the fifth social media platform, Tiktok, the respondents' enjoyment was collected in the table, Table 3.4.3.1. There are 52 more than 120 respondents who enjoyed using Tiktok as a social media platform. Otherwise, approximately 68 out of 120 respondents stated that Tiktok is not a fun social media platform. It is clear that the respondents who did not enjoy Tiktok outnumber the respondents who did enjoy Tiktok. The pie chart in Figure 3.4.3.5 shows the percentage of both "No" and "Yes" responses. The percentage of respondents who say "Yes" is 43.44%, while the percentage of those who say "No" is 56.67%. To summarise, the respondents we surveyed did not enjoy Tiktok as a social media platform.

4.0 CONCLUSION

Our research has concluded with 120 respondents from the Faculty of Computing, University of Technology Malaysia, consisting of 60 males and 60 females. The mode age among our respondents lies at the age of 20, which is our initial target users of social media. The survey we prepared asks the respondents questions revolving around the influence of social media in their daily life. Questions such as their sleep time, type of social media they use, and amount of times they open social media in a day, etc are among the important questions that allow us to make deductions and generalize the influence of social media to undergraduates' everyday.

Based on our findings, 114 out of 120 respondents mainly use Instagram, being the social media with the highest usage, followed by Youtube, Facebook, Tiktok, and finally Twitter. While Youtube and Tiktok could be especially time-consuming, Instagram's features such as notes and stories only require less than a minute to access and can provide equal amounts of entertainment as other social media. It could also be due to geographical conditions that caused more users to prefer Instagram, Facebook, and Youtube over Tiktok and Twitter (country-wise).

In our next findings, we found out a similar amount of people spend their time on social media 3 to 5 hours and 6 to 8 hours respectively. This is quite unexpected as we consider spending more than 5 hours on social media everyday immoderate and may delay educational progression. This can be seen from their sleeping habits being close to 12.00 a.m, and social media being one the possible contributors to such late time of sleeping. Alas, it is also clear that most students require social media for entertainment purposes, as it is one of the main sources of stress reliever from all academic work. 100 respondents agree that the existence of social media provides benefits for them, while 42 respondents also agree that they are harmful to them, yet most students (56 respondents) stay neutral on the discussion on whether social media is harmful. While social media is able to give them the comfort of relaxation, uncontrolled usage of social media can lead to addiction and laziness, which could eventually even lead to sedentary lifestyles.

To sum up our research, we would like to bring forward the vast amount of knowledge we gained from conducting our survey. In the early phase of data collecting, we learned about how to create Google Forms effectively and efficiently to ask questions to our target population with ease. We next gained the idea on how to spread our survey around to

maximize the number of respondents by sharing the link on social media specifically Whatsapp and Telegram where we can get contact with students of the Faculty of Computing easily. Next, we were able to learn about how to create statistical charts using R programming. By using R programming, creating graphs such as bar charts, box plots, or histogram only needs a few lines of coding. Not only that, it can make calculations such as mean, quartiles, etc at an instant. This helps accelerate our data analysis process as we can easily see from the charts regarding the data we collected thoroughly.

We believe the conduction of this research will assist us in our future in terms of both technical skills and communication skills, allowing us to be an efficient worker in future companies if data collection tasks are given to us. We also believe that by putting our hands on experimenting the entire process of collection and analysis of data, we can communicate with our work partners easily about tasks that even we might not be in charge of. Finally, we want to thank our lecturer, Dr Noorfa Haszlinna binti Mustaffa for guiding us on our project and our 120 respondents for their participation in helping us complete our research.

Screenshot of Google Form

QuestionsResponses120Settings



Section 1 of 9

Social Media and it's Influence in our Daily Life

Dear Respondents,

Salam Sejahtera, 您好, Have A Great Day, வணக்கம்

We are the students from Faculty of Computing, Universiti Teknologi Malaysia.

Thank you for spending the time to take part in this survey for our Probability & Statistical Data Analysis course **Assignment 1** which titled The Usage of Social Media among students of Faculty Computing, UTM

You may take approximately 5-10 minutes to finish three relevant parts about 15 Questions. **All data and answers will be kept secret and use for survey purposes.**

Presented By:

1. LIM JING YONG A22EC0182
2. MARCUS JOEY SAYNER A22EC0193
3. SOH HAN WEI A22EC0273

+

📄

Tt

🖼️

▶

☰

Your social media usage

Description (optional)

How many of these social media applications do you currently use? *

☐ Youtube

☐ Facebook

☐ Instagram

☐ Twitter

☐ Tiktok

☐ None of the above

What's the total of hours you spent on your phone on a daily basis? *

☐ 0 - 2 hours

☐ 3 - 5 hours

☐ 6 - 8 hours

☐ 9 - 11 hours

Age *

Short-answer text

Gender *

☐ Male

☐ Femole

Current occupation *

☐ Student

☐ Worker

☐ Unemployed

After section 1 Continue to next section

What time do you normally sleep? (24H FORMAT) *

Time 

In your opinion, when do you think is the most optimum time to sleep? (24H FORMAT) *

Time 

After section 2 Continue to next section ▼

Section 3 of 9

Your opinion on social media  

Description (optional)

Do you use social media for entertainment purposes? *

☐ Yes

☐ No



Section 9 of 9

Thank you for taking your time to answer our google form! Much appreciation for your help  

Description (optional)

Image title



Why do you use this social media platform?

☐ Entertainment

☐ Education

☐ Other

How often do you open this social media platform in one day?

Short-answer text

Do you enjoy using this social media platform?

☐ Yes

☐ No